W03 报告

1. 测试范围与环境

本文核对的是向量探针、NQ 代理评测、COCO 图像到 caption，以及 `packages/core_embedding` 的默认单测和行覆盖率。

系统环境：

- Dart 3.13.4（stable）
- Flutter 3.47.5，`D:\flutter`
- Java 17.0.4
- Python 3.13.2，onnxruntime 1.30.0，tokenizers 0.23.2，pillow 12.3.0，numpy 2.5.3

2. 用例表

用例 ID	命令	期望	实际	判定	证据
EMB-01	`packages\core_embedding` 下 `dart analyze`	0 issue	`No issues found!`	通过	`logs/dart_analyze.txt`；`screenshots/EMB-01-dart-analyze.png`
EMB-02	同目录 `dart test --exclude-tags integration`	默认套件全绿，不加载模型	`00:03 +93: All tests passed!`	通过	`logs/dart_test.txt`；`screenshots/EMB-02-dart-test.png`
EMB-03	同目录覆盖率两条命令，`--report-on=lib`	行覆盖率 ≥ 85.00%	1250/1458 = 85.73%。带覆盖率的那次结束行是 `00:15 +93: All tests passed!`	通过	`logs/coverage.txt`；`coverage/lcov.info`；覆盖率报告.md；`screenshots/EMB-03-coverage.png`
EMB-04	`dart run packages\core_embedding\bin\embed_probe.dart --path datasets\samples`	扫描、解析、切块、出向量；打印结果、维度、耗时和前 3 维	入库 3 条全 ok，7690 ms。向量 97 条全 ok，0 失败，212987 ms。文档文本 96 条、384 维；`sample.png` 1 条、512 维、`clipShared`。DOCX 22 条页码为空，PDF 第 1 页 37 条、第 2 页 37 条	通过	`logs/embed_probe_samples.txt`；`screenshots/EMB-04-embed-probe-samples.png`
EMB-05	`dart run packages\core_embedding\bin\embed_probe.dart --path datasets\coco\val_sample\images`	50 张图各出一条图像向量	入库 50 条全 ok，1848 ms。向量 50 条全 ok，78466 ms，维度都是 512，空间都是 `clipShared`	通过	`logs/embed_probe_coco.txt`；`screenshots/EMB-05-embed-probe-coco.png`
EMB-06	`engine\.venv\Scripts\python.exe engine\eval_nq.py`	300 题打出 recall@1、recall@5、MRR，并标明是代理指标	recall@1 = 0.110000（33/300），recall@5 = 0.210000（63/300），MRR = 0.166034。日志首行是 `metric_layer=embedding_proxy`	通过	`logs/nq_eval.txt`；模型精度验证报告.md 第 4 节；`screenshots/EMB-06-nq-eval.png`
EMB-07	`engine\.venv\Scripts\python.exe engine\eval_coco.py`	有文本塔则报 image→caption recall@1、recall@5	文本塔 fp32 存在。recall@1 = 0.920000（46/50），recall@5 = 1.000000（50/50），平均排名 1.14。`function_status=complete`	通过	`logs/coco_eval.txt`；模型精度验证报告.md 第 5 节；`screenshots/EMB-07-coco-eval.png`
EMB-08	证据集文件与说明	≥ 20 条，dev/holdout 分开，写下冻结时间，含跨页题，本周不算检索分	25 条，dev 15，holdout 10。冻结于 2026-10-16T13:12:33Z。`ev-08`、`ev-16`、`ev-17` 为跨页。`scored=0`	通过	`evidence_set.jsonl`；证据集说明.md；`logs/evidence_freeze.txt`
EMB-09	`apps\desktop` 下 `flutter pub get --offline`	桌面端仍能解析对 `core_embedding` 的 path 依赖	退出码 0。lock 新增传递依赖 `unorm_dart` 0.2.0	通过	`logs/flutter_pub_get.txt`
INT-01	`packages\core_embedding` 下 `dart test --tags integration`	真模型套件全绿（加载 tokenizer.json、	`00:15 +7: All tests passed!`	通过	`screenshots/INT-01-dart-test-integration.png`

用例 ID	命令	期望	实际	判定	证据
		fp32 ONNX、数据集图片）			
A3-01	`engine\.venv\Scripts\python.exe engine\embed_spike.py`	一句中文、一句英文、一张 COCO 图各出一次向量	中文 384 维 52.86 ms、英文 384 维 55.54 ms、COCO 图 512 维 60.10 ms，`finite=1`，中文 `unk_tokens=0`	通过	`logs/embed_spike.txt`；`screenshots/A3-embed-spike.png`
A3-02	`engine\.venv\Scripts\python.exe engine\convert_check.py`	转换前后余弦 ≥ 0.999	三份 fp32 → `ORT_ENABLE_ALL` 最低余弦 1.000000；三份 int8 为 0.982951 / 0.045810 / 0.768964，均不采用	通过	`logs/convert_check.txt`； ADR 0003 决策 5； `screenshots/A3-convert-check.png`

3. 交付物

编号	文件	状态
G1 / G5	`packages/core_embedding/bin/embed_probe.dart`	已完成。EMB-04 出向量 97 条、EMB-05 出向量 50 条，失败 0
G2	`reports/week3/模型精度验证报告.md`	已完成。7 节，四层指标分列
G3	`reports/week3/coverage/lcov.info`、`覆盖率报告.md`	已完成。1250/1458 = 85.73%，门槛 85.00%
G4	`docs/api/embedding-module-api.md`	已完成。53 个 public 符号小节
G6	`evidence_set.jsonl`、`证据集说明.md`	已完成。25 条，dev 15 / holdout 10，`scored=0`
G7	ADR 0003、TDD Q1/Q3/Q7	已完成。三条均标记已结案
G8	本报告、`logs/`、`screenshots/`	已完成。12 条用例

4. 证据索引

路径	对应内容
`reports/week3/logs/dart_analyze.txt`	EMB-01
`reports/week3/logs/dart_test.txt`	EMB-02
`reports/week3/logs/dart_test_coverage_run.txt`	EMB-03 带 `--coverage` 的那次套件输出
`reports/week3/logs/coverage.txt`	EMB-03 命令、行命中与分文件数字
`reports/week3/logs/embed_probe_samples.txt`	EMB-04
`reports/week3/logs/embed_probe_coco.txt`	EMB-05
`reports/week3/logs/embed_probe.txt`	EMB-04 与 EMB-05 接在一起的副本，每段有 `# run=`
`reports/week3/logs/nq_eval.txt`	EMB-06
`reports/week3/logs/coco_eval.txt`	EMB-07
`reports/week3/logs/evidence_freeze.txt`	EMB-08 冻结记录
`reports/week3/logs/flutter_pub_get.txt`	EMB-09
`reports/week3/logs/embed_spike.txt`	A3-01
`reports/week3/logs/convert_check.txt`	A3-02
`reports/week3/logs/text_coverage_b.txt`	阶段 B 的覆盖率（494/719 = 68.71%）
`reports/week3/logs/text_coverage_c.txt`	阶段 C 与 D 的覆盖率（1250/1458 = 85.73%）
`reports/week3/logs/chunk_chars_calibration.txt`	`chunkChars=120` 在 `datasets/samples` 上的标定
`reports/week3/logs/image_preprocess_pillow.txt`	纯 Dart 双线性缩放对 Pillow 参考的对照
`reports/week3/coverage/lcov.info`	G3 交付物
`reports/week3/evidence_set.jsonl`	G6 自建证据集
`reports/week3/screenshots/*.png`	第 2 节各条命令的终端截屏
`docs/api/embedding-module-api.md`	G4
`docs/architecture/adr/0003-embedding-model-and-runtime.md`	G7

5. 截屏

按第 2 节的用例 ID 排列。

```
accessible-multimodal-retrieval on  feat/w3-embedding-validation
❯ dart analyze
Analyzing accessible-multimodal-retrieval...
No issues found!
```

```
accessible-multimodal-retrieval\packages\core_embedding on  feat/w3-embedding-validation [?] is  v0.1.0 via
❯ dart test --exclude-tags integration
00:03 +93: All tests passed!
```

```
accessible-multimodal-retrieval\packages\core_embedding on  feat/w3-embedding-validation [?] is  v0.1.0 via  took 8s
❯ dart test --coverage=coverage --exclude-tags integration
00:15 +93: All tests passed!

accessible-multimodal-retrieval\packages\core_embedding on  feat/w3-embedding-validation [?] is  v0.1.0 via  took 20s
❯ dart run coverage:format_coverage --lcov --in=coverage --out=coverage/lcov.info --report-on=lib --package=.

accessible-multimodal-retrieval\packages\core_embedding on  feat/w3-embedding-validation [?] is  v0.1.0 via  took 7s
❯ $lf = 0; $lh = 0
· Get-Content coverage\lcov.info | ForEach-Object {
·   if ($_ -match '^LF:(\d+)') { $lf += [int]$Matches[1] }
·   if ($_ -match '^LH:(\d+)') { $lh += [int]$Matches[1] }
· }
· "lines hit $lh / $lf = $([math]::Round(100.0 * $lh / $lf, 2))%"
lines hit 1250 / 1458 = 85.73%
```

```
accessible-multimodal-retrieval on  feat/w3-embedding-validation [?]
❯ dart run packages\core_embedding\bin\embed_probe.dart --path datasets\samples
progress 0/3
progress 0/3 sample.docx
progress 1/3 sample.docx
progress 1/3 sample.pdf
progress 2/3 sample.pdf
progress 2/3 sample.png
progress 3/3 sample.png
root: E:\Project\accessible-multimodal-retrieval\datasets\samples
embed_ms_includes=process_start_and_model_load
text_model=paraphrase-multilingual-MiniLM-L12-v2  image_model=mobileclip-s0-vision
ingest_total: 3  ingest_ok: 3  ingest_partial: 0  ingest_skipped: 0  ingest_failed: 0  ingest_cancelled: false  ingest_ms: 7
690
embed_total: 97  embed_ok: 97  embed_failed: 0  embed_cancelled: false  embed_ms: 212987  embed_calls: 97
```

```
embed_ms_includes=process_start_and_model_load
text_model=paraphrase-multilingual-MiniLM-L12-v2  image_model=mobileclip-s0-vision
ingest_total: 50  ingest_ok: 50  ingest_partial: 0  ingest_skipped: 0  ingest_failed: 0  ingest_cancelled: false  ingest_ms:
 1848
```

```
accessible-multimodal-retrieval on  feat/w3-embedding-validation [?] took 23s
❯ engine\.venv\Scripts\python.exe engine\eval_nq.py
metric_layer=embedding_proxy
not_measured=document_retrieval
not_measured=generation
protocol=nq-a
query_order=nq_open_val50 then nq_open_val250_part2
pool=600
pool_parts=first_gold_answer:300 question_distractor:300
ranking_exclusion=drop the query's own question row; the other 299 questions stay
ranking_pool=599
relevant=the paired first gold answer row
tie_break=higher cosine, then lower candidate index
tokenizer=tokenizers no_truncation manual_pad
pooling=mean attention_mask
normalized=l2
max_length=128
model=models/raw/text/paraphrase-multilingual-MiniLM-L12-v2/onnx/model.onnx
nq_file=datasets/nq/validation/nq_open_val50.jsonl lines=50
nq_file=datasets/nq/validation/nq_open_val250_part2.jsonl lines=250
load_ms=932.29
embedded=600 dim=384 space=documentText
infer_ms_median=34.02
infer_ms_max=79.32
infer_ms_excludes=model_load
queries_scored=300
queries_skipped=0
recall@1_hits=33/300
recall@5_hits=63/300
recall@1=0.110000
recall@5=0.210000
mrr=0.166034
rank_min=1
rank_median=47.0
rank_max=526
chance_recall@1=0.001669
chance_recall@5=0.008347
single_relevant=1
recall_equals_hit=1
example index=0 rank=11 question=when was the last time anyone was on the moon gold=14 December 1972 UTC
example index=1 rank=82 question=who wrote he ain't heavy he's my brother lyrics gold=Bobby Scott
example index=2 rank=288 question=how many seasons of the bastard executioner are there gold=one
gate=pass
```

```
accessible-multimodal-retrieval on  feat/w3-embedding-validation [?] took 24s
❯ engine\.venv\Scripts\python.exe engine\eval_coco.py
metric_layer=image_caption_sanity
not_measured=pdf_figure_understanding
not_measured=statistical_precision
space=clipShared
caption_encoder=mobileclip-s0-text
caption_encoder_not=paraphrase-multilingual-MiniLM-L12-v2
n_captioned_images=50
n_rvlcdip_unlabeled=16
rvlcdip_in_this_run=0
captions=datasets/coco/val_sample/captions.jsonl lines=50
vision_model=models/raw/image/mobileclip-s0/onnx/vision_model.onnx
text_model=models/raw/image/mobileclip-s0/onnx/text_model.onnx
preprocess=shortest_edge_256 bilinear center_crop rescale_1_255 do_normalize=false
text_pad=exclamation_to_77
max_length=77
load_ms=468.29
embedded_images=50 dim=512
embedded_captions=50 dim=512
image_infer_ms_median=77.83
image_infer_ms_max=142.81
text_infer_ms_median=22.65
text_infer_ms_max=87.29
infer_ms_excludes=model_load
block=image_to_caption
function_status=complete
recall@1_hits=46/50
recall@5_hits=50/50
recall@1=0.920000
recall@5=1.000000
mean_rank=1.14
rank_min=1
rank_median=1.0
rank_max=4
chance_recall@1=0.020000
chance_recall@5=0.100000
example index=0 file=000000397133.jpg rank=1 caption=A man is in a kitchen making pizzas.
example index=1 file=000000037777.jpg rank=1 caption=The dining table near the kitchen has a bowl of fruit on it.
example index=2 file=000000252219.jpg rank=1 caption=a person with a shopping cart on a city street
block=image_to_image_sanity
image_to_image_not_completion=1
self_cosine_min=0.999999
self_cosine_max=1.000000
nearest_other_cosine_median=0.510716
nearest_other_cosine_max=0.826031
gate=pass
```

```
accessible-multimodal-retrieval\packages\core_embedding on  feat/w3-embedding-validation [?] is 📦 v0.1.0 via 🎯
❯ dart test --tags integration
00:15 +7: All tests passed!
```

```
accessible-multimodal-retrieval on  feat/w3-embedding-validation [?] took 7s
❯     engine\.venv\Scripts\python.exe engine\embed_spike.py
runtime=onnxruntime-cpu
onnxruntime=1.30.0
tokenizers=0.23.2
pillow=12.3.0
numpy=2.5.3
infer_ms=mean of 3 runs after 1 warmup; session graph optimization disabled
text_model=paraphrase-multilingual-MiniLM-L12-v2
text_file=E:\Project\accessible-multimodal-retrieval\models\raw\text\paraphrase-multilingual-MiniLM-L12-v2\onnx\model.onnx
image_model=mobileclip-s0
image_file=E:\Project\accessible-multimodal-retrieval\models\raw\image\mobileclip-s0\onnx\vision_model.onnx
image_preprocess=shortest_edge_256,bilinear,center_crop_256,rescale_1_255,do_normalize=false
text_input name=input_ids type=tensor(int64) shape=['batch_size', 'sequence_length']
text_input name=attention_mask type=tensor(int64) shape=['batch_size', 'sequence_length']
text_input name=token_type_ids type=tensor(int64) shape=['batch_size', 'sequence_length']
text_output name=last_hidden_state type=tensor(float) shape=['batch_size', 'sequence_length', 384]
sentence index=0 lang=zh dim=384 l2_raw=3.823005 l2_normalized=1.000000 infer_ms=52.86 content_tokens=13 unk_tokens=0 finite
=1
sentence index=1 lang=en dim=384 l2_raw=4.611503 l2_normalized=1.000000 infer_ms=55.54 content_tokens=13 unk_tokens=0 finite
=1
image_input name=pixel_values type=tensor(float) shape=['batch_size', 3, 256, 256]
image_output name=image_embeds type=tensor(float) shape=['batch_size', 512]
image file=000000397133.jpg dim=512 l2_raw=0.821738 l2_normalized=1.000000 infer_ms=60.10 finite=1
gate=pass
```

```
clip_text_ort_cosine_min=1.000000
clip_text_conversion=ort_enable_all
clip_text_conversion_cosine_min=1.000000
clip_text_shipped=fp32
clip_text_int8_rejected cosine_min=0.768964 threshold=0.999
clip_text_gate=pass
ort_optimize_note=ORT_ENABLE_ALL may embed hardware-specific kernels; cosine 1.0 means the original fp32 file stays the port
able artifact
overall_gate=pass
```